

Introduction to Robotics

Personal Notes

Mihir Rao

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1 Intro to Robotics

1.1 Motivation and scope

Robotics brings computation into contact with the physical world. The opening examples range from household manipulation and humanoid locomotion to autonomous driving and drone delivery. They motivate a common goal: machines that perform useful tasks reliably outside a controlled demonstration.

The course develops the mathematical and algorithmic foundations of these systems, connecting machine learning, control, optimization, and computer vision. Deployment also raises questions about privacy, employment, safety, and whether a machine's behavior matches human intentions. These concerns accompany the technical problems rather than disappearing when a demonstration succeeds.

1.2 What counts as a robot?

The lecture places modern robotics in a longer history of imagined artificial beings and mechanical automata. Two landmarks are Karel Čapek's 1920 play *R.U.R.*, which introduced the word *robot*, and the development of the Unimate industrial manipulator by George Devol and Joseph Engelberger.

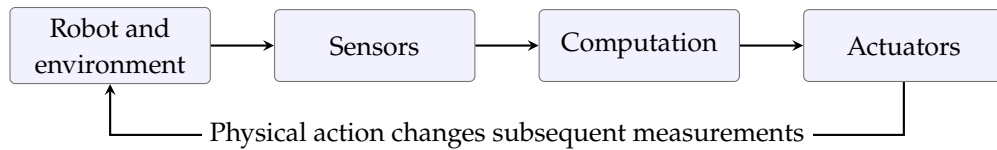
Definition: A working definition of a *robot* is a programmable physical agent whose behavior accomplishes tasks in its environment. Here, *embodiment* means that the agent has a physical form through which it interacts with the world.

Different definitions emphasize different properties: automatic execution of actions, reprogrammable manipulation, or the combination of sensing, computation, and physical action. These choices change which systems qualify. A dishwasher and an automatic door test how broad a definition should be; a surgical system controlled by a person tests whether autonomy is essential. A humanoid shape is not required, and programmability does not by itself imply independence from human supervision.

Note. There is no universally sharp boundary between robots and other machines. The working definition helps organize the subject; it is not a test that settles every borderline example.

1.3 The sensing–computation–action loop

A robotic system combines three functions. A *sensor* measures aspects of the robot or its surroundings. *Computation* converts measurements and task information into decisions. An *actuator* turns commands into physical effects, such as the thrust produced by a drone's motor-driven rotors. The resulting motion changes what the sensors observe next.



Original schematic of the lecture’s sensing–computation–action cycle.

The lecture uses a small quadrotor as a concrete example: its sensors include an inertial measurement unit and an optical-flow camera, its onboard computation processes information, and its rotors provide actuation. A *feedback* loop uses new measurements to adjust subsequent actions.

1.4 Three connected technical areas

Planning selects a course of action toward a goal; *control* determines how to execute and adjust that action. *Geometric motion planning* focuses on the spatial feasibility of motion, including obstacle avoidance. *Dynamics* describes how the robot’s motion evolves under applied inputs. The course moves from geometric planning to robot dynamics, feedback control, and planning subject to those dynamics.

State estimation infers quantities describing the robot from available measurements. *Localization* estimates where the robot is, while *mapping* estimates the surrounding environment. The course introduces Bayes filtering, Kalman filters, and particle filters as estimation methods. *Simultaneous localization and mapping (SLAM)* addresses localization and map construction together.

Robot learning uses data or experience to improve robotic behavior. The roadmap includes imitation learning, generative architectures, reinforcement learning, and world models. These are introduced here as topics for later lectures, not yet as algorithms with specified assumptions or guarantees.

1.5 Uncertainty as a common obstacle

Uncertainty is incomplete or imperfect knowledge relevant to a decision. The lecture connects it to all three technical areas:

Area	Sources of uncertainty
Planning and control	Robot dynamics, changes in the world, and the robot’s initial condition.
Localization and mapping	The robot’s state, environmental geometry, and the meaning of objects or observations.
Robot learning	Data quality, human intent, and the rewards supplied by interaction with the world.

Example: For a drone navigating around obstacles, a proposed route is useful only if the drone can execute it and estimate where it is. An imperfect motion model affects execution, while an inaccurate position estimate affects both control and obstacle avoidance. This illustrates why the three areas must work together.

1.6 Learning through implementation

The hardware platform is the Crazyflie 2.1 brushless quadrotor, a small, lightweight, open-source drone. The practical objective is autonomous navigation. Hardware work exposes the gap between a mathematical design and its behavior under real sensing and actuation. The lecture organizes this work in teams, with testing in designated netted spaces after the required lab training.

The assumed background is multivariable calculus, linear algebra, elementary probability, differential equations, and programming; the course uses Python. In the presented offering, problem sets account for 40% of the grade, the in-person midterm for 30%, and the final project for 30%. Assignments combine theory, code, and hardware. Collaboration is encouraged, but individual submissions remain the rule outside the team hardware portions.

The lecture's AI policy permits reviewing past assignments, explanations of technical concepts, Python syntax questions, and unrestricted use for hardware assignments and the final project. Its educational point is to develop skill with these tools while retaining an independent command of the fundamentals. Current administrative details belong to the course syllabus and Canvas.

There is no required textbook. Suggested references are *Principles of Robot Autonomy*, *Planning Algorithms*, and *Probabilistic Robotics*. ROB 549 is the graduate track and includes additional assignment problems. The technical sequence begins with geometric planning.

Source: Anirudha Majumdar, *Introduction to Robotics*, Princeton University, ROB 345 / 549, Lecture 1, "Introduction to Robotics," [lecture slides](#), PDF pages 1–57. [Official course materials](#); accessed September 19, 2026. These notes paraphrase the slides; the schematic and drone example are original explanatory additions.